UNITED STATES
NUCLEAR REGULATORY COMMISSION
REGION III
2056 WESTINGS AVENUE, SUITE 400
NAPERVILLE, IL 60563-2657

March 3, 2026

EAF-RIII-2025-0230
EN 57208
NMED No. 240233 (Closed)

Phil Leeker
Radiation Safety Officer
Midwest Subsurface Testing, LLC
8606 Page Ave.
St. Louis, MO 63114

SUBJECT: NRC REACTIVE INSPECTION REPORT NO. 03032036/2024001(DRSS) AND NOTICE OF VIOLATION – MIDWEST SUBSURFACE TESTING, LLC

Dear Phil Leeker:

This letter refers to a reactive inspection conducted on July 18, 2024, through August 8, 2024, by an inspector from the U.S. Nuclear Regulatory Commission (NRC) conducted remotely, with continued in-office review through February 24, 2026. The purpose of the inspection was to review activities performed under your NRC license to ensure that these activities were being performed in accordance with NRC requirements. The purpose of the in-office review was to evaluate additional information not available during the on-site inspection. Daniel Fiedorek of my staff conducted a final exit meeting by telephone with you and Dan Barczykowski on February 24, 2026, to discuss the inspection findings. This letter presents the results of the inspection.

This inspection examined activities conducted under your license as they relate to public health and safety, and to confirm compliance with the NRC's rules and regulations, and with the conditions of your license. Within these areas, the inspection consisted of selected examination of procedures and representative records, observations of activities, and interviews with personnel.

Based on the results of this inspection, the NRC has determined that one Severity Level IV violation of NRC requirements occurred. The violation concerned the failure to maintain control of licensed material in use at a temporary jobsite as required by Title 10 of the *Code of Federal Regulations* (10 CFR) 20.1802. The violation was evaluated in accordance with the NRC Enforcement Policy, available on the NRC's website at http://www.nrc.gov/about-nrc/regulatory/enforcement/enforce-pol.html. The violation is cited in the enclosed Notice of Violation (Enclosure 1). The NRC is citing the violation in the enclosed Notice because the inspector identified it.

The NRC has concluded that information regarding the reason for the violation, the corrective actions taken and planned to correct the violation and prevent recurrence, and the date when full compliance will be achieved is already adequately addressed on the docket in this letter in the enclosed inspection report (Enclosure 2). Therefore, you are not required to respond to this letter unless the description herein does not accurately reflect your corrective actions or your

position. In that case, or if you choose to provide additional information, you should follow the instructions specified in the enclosed Notice.

In accordance with the NRC's "Agency Rules of Practice and Procedure" in 10 CFR 2.390, a copy of this letter, its enclosure and any response you provide will be made available electronically for public inspection in the NRC's Public Document Room or from the NRC's Agencywide Documents Access and Management System (ADAMS), accessible at http://www.nrc.gov/reading-rm/adams.html. To the extent possible, any response you provide should not include personal privacy, proprietary, or security-related information so that it can be made publicly available without redaction.

Please feel free to contact Daniel Fiedorek of my staff if you have any questions regarding this inspection. Mr. Fiedorek can be reached at 630-829-9836 or Daniel.Fiedorek@nrc.gov.

Sincerely,

Signed by Edwards, Rhex
on 03/03/26

Rhex Edwards, Chief
Materials Inspection Branch
Division of Radiological Safety and Security

Docket No. 030-32036
License No. 24-24619-02

Enclosure:
1. Notice of Violation
2. Inspection Report No. 03032036/2024001(DRSS)

cc (w/encl): Dan Barczykowski,
Principal
State of Missouri

Letter to P. Leeker from R. Edwards, dated March 3, 2026.

SUBJECT: NRC REACTIVE INSPECTION REPORT NO. 03032036/2024001(DRSS) AND NOTICE OF VIOLATION – MIDWEST SUBSURFACE TESTING, LLC

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ADAMS Accession Number: ML26061A232

OFFICE	RIII-DRSS		RIII-EICS		RIII-DRSS			
NAME	DFiedorek:mh		DBetancourt-Roldan		REdwards			
DATE	3/03/2026		3/03/2026		3/03/2026			

OFFICIAL RECORD COPY

NOTICE OF VIOLATION

Midwest Subsurface Testing, LLC
St. Louis, MO

License No. 24-24619-02
Docket No. 030-32036
EAF-RIII-2025-0230

During a U.S. Nuclear Regulatory Commission (NRC) inspection conducted on July 18, 2024 through August 8, 2024, with continued in-office review through February 24, 2026, one violation of NRC requirements was identified. In accordance with the NRC Enforcement Policy, the violation is listed below:

> Title 10 of the *Code of Federal Regulations* (10 CFR) Part 20.1802 requires that licensees control and maintain constant surveillance of licensed material that is in a controlled or unrestricted area and that is not in storage.
>
> Contrary to the above, On July 3, 2024, the licensee failed control and maintain constant surveillance of license material that was in an unrestricted area and not in storage. Specifically, the licensee failed to control a CPN MC1 Elite portable gauging device, nominally containing 10 mCi Cs-137 and 50 mCi of Am-241/Be, at a temporary job site, resulting in it being damaged by a skid steer.
>
> This is a Severity Level IV violation (Section 6.3).

The NRC has concluded that information regarding the reason for the violation, the corrective actions taken and planned to correct the violation and prevent recurrence, and the date when full compliance was achieved, is already adequately addressed on the docket in this letter in the enclosed inspection report (Enclosure 2). However, you are required to submit a written statement or explanation pursuant to 10 CFR 2.201 if the description therein does not accurately reflect your corrective actions or your position. In that case, or if you choose to respond, clearly mark your response as a "Reply to a Notice of Violation, IR 03032036/2024001(DRSS)" and send it to the U.S. Nuclear Regulatory Commission, ATTN.: Document Control Desk, Washington, DC 20555-0001 with a copy to the Regional Administrator, Region III, within 30 days of the date of the letter transmitting this Notice. Alternatively, you may provide a response electronically to RidsRgn3DRSS.Resource@nrc.gov within 30 days of the date of this letter.

Any response you provide will be made available electronically for public inspection in the NRC Public Document Room or from the NRC's Agencywide Documents Access and Management System (ADAMS), accessible at http://www.nrc.gov/reading-rm/adams.html. Therefore, to the extent possible, your response should not include any personal privacy, proprietary, or security-related information so that it can be made available to the Public without redaction.

If you contest this enforcement action, you should also provide a copy of your response, with the basis for your denial, to the Director, Office of Enforcement, U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001.

In accordance with 10 CFR 19.11, you may be required to post this Notice within two working days of receipt.

Dated this 3rd day of March 2026.

U.S. Nuclear Regulatory Commission
Region III

Docket No.	030-32036
License No.	24-24619-02
Report No.	03032036/2024001(DRSS)
EA No./NMED No.	EAF-RIII-2025-0230 / 240233
Licensee:	Midwest Subsurface Testing, LLC
Facility:	8606 Page Ave. St. Louis, MO
Inspection Dates:	July 18, 2024 – August 8, 2024
Exit Meeting Date:	February 24, 2026
Inspector:	Ryan Craffey, Senior Health Physicist Daniel Fiedorek, Health Physicist
Approved By:	Rhex Edwards, Chief Materials Inspection Branch Division of Radiological Safety and Security

Executive Summary

Midwest Subsurface Testing, LLC
NRC Inspection Report 03032036/2024001

This was an announced reactive inspection of a geotechnical engineering firm authorized by NRC License No. 24-24619-02 to possess portable nuclear density gauges at its facility in St. Louis, Missouri, and use them at temporary jobsites in areas of NRC jurisdiction. The inspection was in response to a reportable event involving damage to a portable gauge containing licensed material at a temporary jobsite in Osage Beach, Missouri, on July 3, 2024.

As a result of this inspection, the NRC identified one Severity Level IV violation of regulatory requirements involving the failure to maintain control of licensed material in use at a temporary jobsite as required by Title 10 of the *Code of Federal Regulations* (CFR) 20.1802.

The findings of the reactive inspection as well as the circumstances surrounding the event, its root causes, contributing factors, and the licensee's corrective actions are discussed in more detail in the following report.

Report Details

1 Program Overview and Inspection History

Midwest Subsurface Testing, LLC is a geotechnical engineering firm authorized to possess and use portable nuclear gauges at its facility in St. Louis, Missouri, and temporary jobsites in NRC jurisdiction. At the time of the inspection, the licensee was authorized to possess 11 Instrotek MC Series Portaprobe gauges and 14 Humboldt Model 5001 gauges.

The NRC previously performed a routine inspection of the licensee on October 26, 2022, and did not identify any violations of NRC requirements. Prior to the inspection in 2022, the NRC performed a routine inspection of the licensee on April 9, 2017, and no violations of NRC requirements were identified.

2 Sequence of Events and Licensee Response

2.1 Inspection Scope

The inspector interviewed licensee personnel to gain a detailed understanding of the circumstances surrounding an event involving damage to a portable gauge and to evaluate the licensee's response to the event.

2.2 Observations and Findings

A. Sequence of Events Leading Up to the Incident

The licensee had a gauge user on site at a temporary jobsite in Osage Beach, Missouri, in the morning of July 3, 2024, to conduct density testing for a retaining wall backfill using a CPN MC1 Elite model InstroTek gauge (serial number 31550) containing nominally 10 mCi of cesium-137 and 50 mCi of americium-241. At approximately 11:44 a.m. Central Daylight Time (CDT), the gauge user momentarily set his equipment down, including the gauge, to get closer to the work to check if they were ready for him to conduct density testing. The equipment was placed in an area where equipment operators were instructed to avoid. In that moment, a skid loader operator incorrectly followed their path and turned toward the gauge. The gauge user attempted to signal the operator to stop but was unable to. The skid loader struck the gauge, and the operator stopped the loader believing he had struck the retaining wall. The gauge user informed the operator that he had struck the gauge and told the operator to park the loader where it was and leave the immediate area but stay on site. The gauge user then contacted the licensee's Radiation Safety Officer (RSO) at approximately 11:46 a.m. CDT.

B. Licensee Response to the Incident

The gauge user immediately informed the RSO that the gauge had been struck at the jobsite. The licensee's RSO informed the gauge user to keep all personnel away from the area until a survey of the gauge could be conducted. The licensee then contacted RM Wester and Associates, based in St. Peters, Missouri, at approximately 12:10 p.m. CDT to perform a site survey and recover the gauge for the licensee.

The RM Wester and Associates technician arrived on site at approximately 2:30 p.m. CDT and found the gauge source rod was broken off at the top of the case. The cesium-137 source remained in the shielded position within the case. The technician took survey readings of the gauge and placed the damaged gauge in its case at approximately 3 p.m. CDT. The technician subsequently performed a survey in the immediate area where the gauge was struck and found no readings above background.

The RM Wester and Associates technician transported the damaged gauge back to their facility. A leak test was performed on July 5, 2024, and indicated that the sources were not leaking.

Title 10 CFR 20.1802 states that the licensee shall control and maintain constant surveillance of licensed material that is in a controlled or unrestricted area and that is not in storage. The gauge user's failure to maintain control and constant surveillance of the portable gauge at the temporary job site in Osage Beach is a violation of 10 CFR 20.1802.

This is an example of a Severity Level IV violation as described in the NRC Enforcement Policy Section 6.3.

As corrective action, the RSO held additional training with the licensee's authorized gauge users to reiterate the necessity of maintaining constant control of gauges at temporary jobsites. This training was conducted by the licensee prior to them submitting their written report.

2.3 Conclusions

The inspector noted one Severity Level IV violation of 10 CFR 20.1802.

3 Notification and Reporting

3.1 Inspection Scope

The inspector reviewed the reports the licensee made to the NRC in response to the event to evaluate compliance with NRC reporting requirements.

3.2 Observations and Findings

Upon learning of the gauge being damaged, the licensee's RSO called the NRC Headquarters Operations Center at 12:30 p.m. CDT to report an event resulting in safety equipment failure in accordance with 10 CFR 30.50(b)(2). Events reported to the NRC under 10 CFR 30.50(b)(2) are required to be reported within 24 hours of an event when equipment is disabled or fails to function as designed. The initial report included a brief description of what occurred, where it occurred, when it occurred, and a description of the licensed material involved in the event. The licensee reported the event approximately 45 minutes after the event occurred.

The licensee submitted a 30-day written report to the NRC on July 18, 2024. The written report contained all information required by 10 CFR 30.50(c)(2) including: a description of the event and its location, details of the equipment and radioactive material involved, corrective actions taken, and confirmation that there were no abnormal exposures to radiation or radioactive material as a result of this event.

3.3 Conclusions

No violations of NRC requirements were identified in reviewing the licensee's notification and reporting of the event.

4 NRC Assessment of the Event

4.1 Inspection Scope

The inspector interviewed licensee personnel and a selection of records to evaluate the consequences of the event and the adequacy of the licensee's response to the event.

4.2 Observations and Findings

A. Root Cause and Contributing Factors

The inspector concluded the root cause of the event was the lack of control of licensed material at a temporary jobsite. Specifically, the root cause of the violation of 10 CFR 20.1802 was an oversight by the gauge user that resulted in the loss of control of material. A contributing factor to the event was human error by the operator of the skid steer who failed to avoid the area where they were instructed to avoid.

B. Independent Assessment of Radiation Exposure

The inspector concluded that since the cesium-137 and americium-241 sources remained in the shielded position at all times and the gauge was able to be safely returned to its transport case, no workers or members of the public were likely to receive measurable exposure to radiation as a result of this event.

C. Assessment of Radiation Safety Practices

The inspector noted that the gauge user followed the licensee's emergency procedures by having the skid loader operator park the skid loader, restricting

access to the gauge, and immediately informing the licensee's RSO of the event. The licensee's RSO also promptly took action by working with a service provider to respond to the event in order to recover the gauge.

4.3 Conclusions

The inspector had no additional findings from the assessment of the event. This event is considered closed.

5 Exit Meeting Summary

The NRC inspector presented their preliminary inspection findings following the remote inspection on August 8, 2024. A final exit was held via videoconference with the licensee on February 24, 2026. The licensee acknowledged the findings presented on both occasions.

LIST OF PERSONNEL CONTACTED

Phil Leeker – Current RSO
Daniel Barczykowski – Principal
Joseph M. Honich – RSO at time of event
Nick Moleski – Authorized User

Attended exit meeting on February 24, 2026

INSPECTION PROCEDURES USED

87103 – Inspection of Nuclear Material Licensees Involved in an Incident or Bankruptcy
87139 – Portable Gauge Programs